



Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)



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WEEKLY LESSON PLAN

Department: Computer Science and Engineering

Year / Semester: 2025-26 / Winter

Course Code / Course Name: 10212CS292 / Applied Programming Skills

Unit III: Hierarchical and Priority-Oriented Data Structures				
Course Outcome: Apply tree and heap structures to handle hierarchical data and priority-based computations.				
Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 5	Session 21	Tree – Tree Traversal	Task 37: Same Tree Task 38: Symmetric Tree	Leet Code
	Session 22		Task 39: Binary Tree Inorder Traversal Task 40: Binary Tree Preorder Traversal Task 41: Binary Tree Postorder Traversal	
	Session 23		Print Birdseye View of a Tree Task 42: Vertical Order Traversal of a Binary Tree	
	Session 24	Path sum Task 43: Path Sum Task 44: Path Sum II		
	Session 25	Root-to-leaf problems Task 45: Binary Tree Paths		

Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 6	Session 26	Diameter/height problems	Task 46: https://takeuforward.org/data-structure/calculate-the-diameter-of-a-binary-tree Task 47: https://takeuforward.org/data-structure/maximum-depth-of-a-binary-tree	takeuforward / Leet Code
	Session 27	Lowest Common Ancestor	Task 48: Lowest Common Ancestor of a Binary Tree Task 49: Lowest Common Ancestor of a Binary Search Tree	
	Session 28	Kth smallest/largest in BST	Task 50: Kth Smallest Element in a BST Task 51: Kth Largest Element in an Array	
	Session 29	Heap Top-K pattern	Task 52: Top K Frequent Elements	
	Session 30	K-way merge using heap	Task 53: Find K Pairs with Smallest Sums	